Install WSL (Skip if you are already using Ubuntu)

```
wsl --install
```

X Server (Optional. Only needed for GUI.)

download and install https://sourceforge.net/projects/vcxsrv/
Start the X server application.

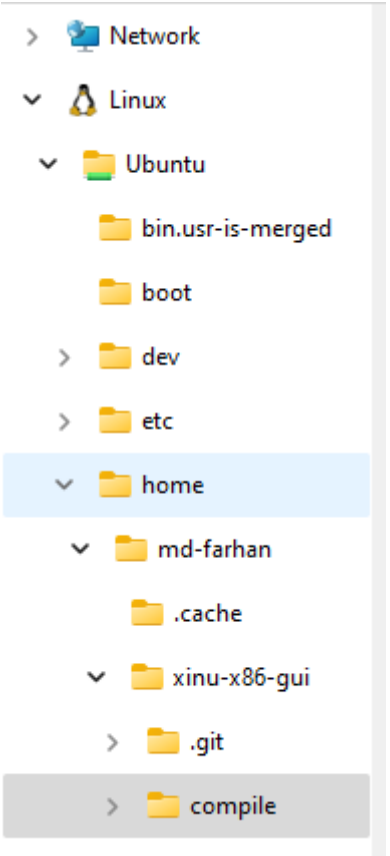
Install QEMU

```
sudo apt update
sudo apt upgrade
sudo apt install qemu-system qemu-utils
sudo apt install flex bison gawk qemu-system-x86 xorriso grub-common
grub-pc-bin
```

Build XINU OS

```
git clone https://github.com/zrafa/xinu-x86-gui
```

Navigate to the `Makefile` using file explorer.
For WSL.
`Linux > Ubuntu > home > username >xinu-x86-gui > compile`
For Ubuntu
`home > username >xinu-x86-gui > compile`



Inside `Makefile`, find `run-qemu` and add this to the end of the command `-nographic`

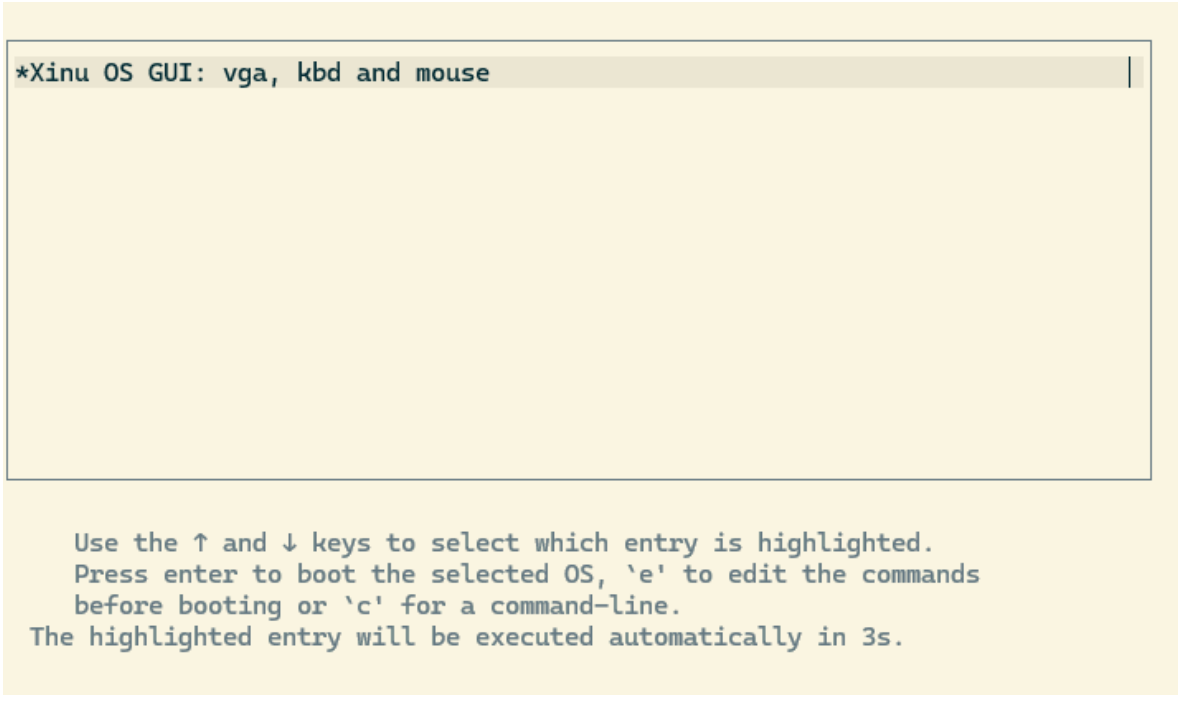
```
172  # Generates the Grub2-based ISO - see bochsrc.txt for info.
173  # Requires grub2-mkrescue and xorriso be installed!
174  run-qemu: qemu-iso-pre
175      echo OTRO GRUB PARA PROBAR: grub-mkrescue -o disk.iso test/img
176      grub-mkrescue -d /usr/lib/grub/i386-pc -o disk.iso test/img
177      qemu-system-i386 -machine accel=kvm -enable-kvm -drive
         file=disk.iso,index=0,media=cdrom -netdev user,id=u0 -device
         e1000-82545em,netdev=u0 -nographic
```

(run once for every time you boot WSL/Ubuntu)

```
sudo chmod 666 /dev/kvm
```

(run every time you boot XINU)

```
cd xinu-x86-gui/compile
make clean
make
make run-qemu
```



```
-----------------------------------------

   __    __   _____   _   _   _    _
   \ \  / /  |__ __|  | \ | | | |  | |
    \ \/ /     | |    |  \| | | |  | |
    / /\ \    _| |_   | \   | | |__| |
   / /  \ \  |_____|  | |\ |  \____/
   --    --   -----   -   -    ----

-----------------------------------------


Welcome to Xinu!

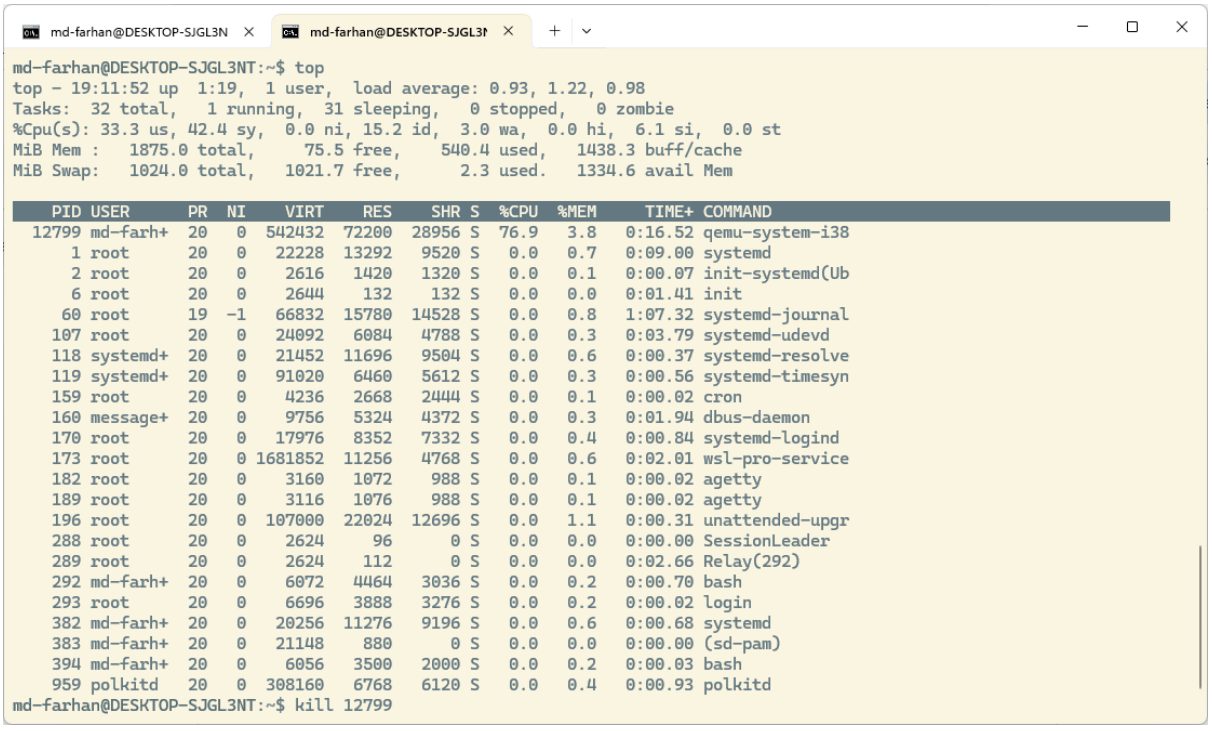

type   help [ENTER]   for ehhhh... help.

xsh $ Obtained IP address  10.0.2.15   (0x0a00020f)
Thu  Jan  9   8:07:46 EST 2025
mouse x: 0, y: 0

xsh $
```

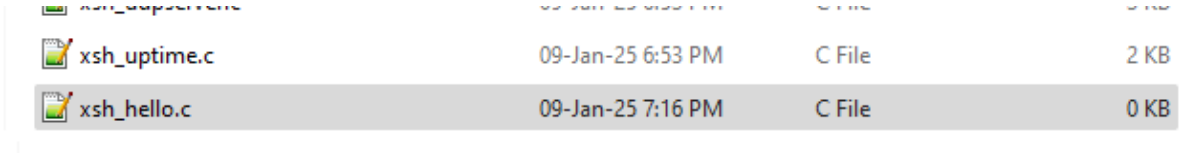
Shut down XINU

Open another terminal. (For WSL, you need to start a new WSL terminal.)
Use the `top` command to see running processes. Get PID of QEMU.
Press q to exit top.
`kill <PID>`



Running code in XINU

Create a c file, `xsh_hello.c`, in `xinu-x86-gui > shell`



Open and write code inside it

```c
#include <xinu.h>
#include <stdio.h>

shellcmd xsh_hello(int nargs, char *args[]) {
    printf("Hello World\n");
    return 0;
}

```

Open `cmdtab.c` from same directory (`xinu-x86-gui > shell`)



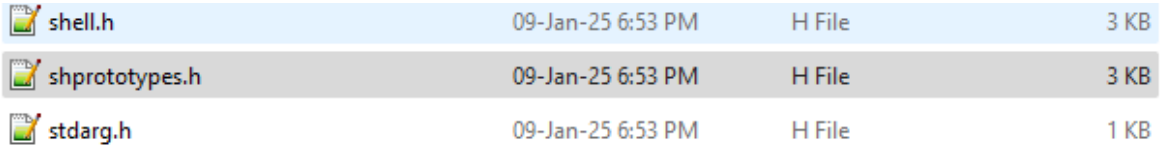
Add new entry

```c
#include <xinu.h>
#include "shprototypes.h"

/************************************************
/* Table of Xinu shell commands and the function ass
/************************************************
const   struct  cmdent  cmdtab[] = {
    {"hello", FALSE, xsh_hello},
    {"argecho", TRUE,   xsh_argecho},
    {"arp",     FALSE,  xsh_arp},
    {"cat",     FALSE,  xsh_cat},
```

(`FALSE`, because `hello` is not a built in command)

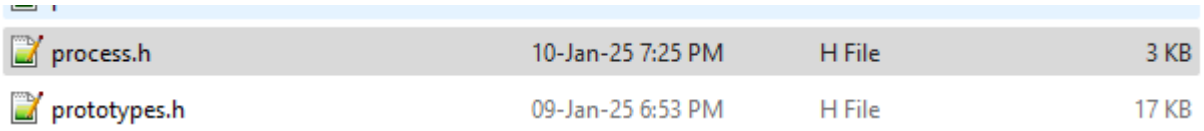
Open `shprototypes.h` header file from `xinu-x86-gui > include`



Add prototype of our function

```c
/* in file xsh_hello.c */
extern  shellcmd  xsh_hello (int32, char *[]);
```

Open `process.h` header file from `xinu-x86-gui > include`



Change `NPROC` from 8 to 20

```
#ifndef NPROC
#define NPROC       20
#endif
```

Build and run the the OS from `xinu-x86-gui > compile` directory

```
cd xinu-x86-gui/compile
make clean
make
make run-qemu
```

In xinu shell, type command `hello` and press enter

```
xsh $
xsh $
xsh $
xsh $ hello
Hello World
xsh $
```

Troubleshooting

This section was contributed by Syed Shifat E Rahman (2003065)

You might encounter grub related errors while running make run-qemu.

```
pc:~/xinu-x86-gui/compile$ make run-qemu
Rebuilding the .defs file

Loading object files to produce GRUB bootable xinu
/usr/bin/ld: warning: binaries/kbddispach.o: missing .note.GNU-stack section implies executabl
e stack
/usr/bin/ld: NOTE: This behaviour is deprecated and will be removed in a future version of the
 linker


        XINU ready.

        make run-qemu    # ...to boot the OS on qemu

rm -rf test/img
mkdir -p test/img/boot/grub
cp xinu.elf test/img/
cp test/grub/grub.cfg test/img/boot/grub
echo OTRO GRUB PARA PROBAR: grub-mkrescue -o disk.iso test/img
OTRO GRUB PARA PROBAR: grub-mkrescue -o disk.iso test/img
grub-mkrescue -d /usr/lib/grub/i386-pc -o disk.iso test/img
grub-mkrescue: error: /usr/lib/grub/i386-pc/modinfo.sh doesn't exist. Please specify --target
or --directory.
make: *** [Makefile:176: run-qemu] Error 1
```

```
sudo apt install grub-pc-bin
sudo apt reinstall grub-pc-bin
grub-mkrescue --target=i386-pc -d /usr/lib/grub/i386-pc -o disk.iso test/img
make run-qemu
```